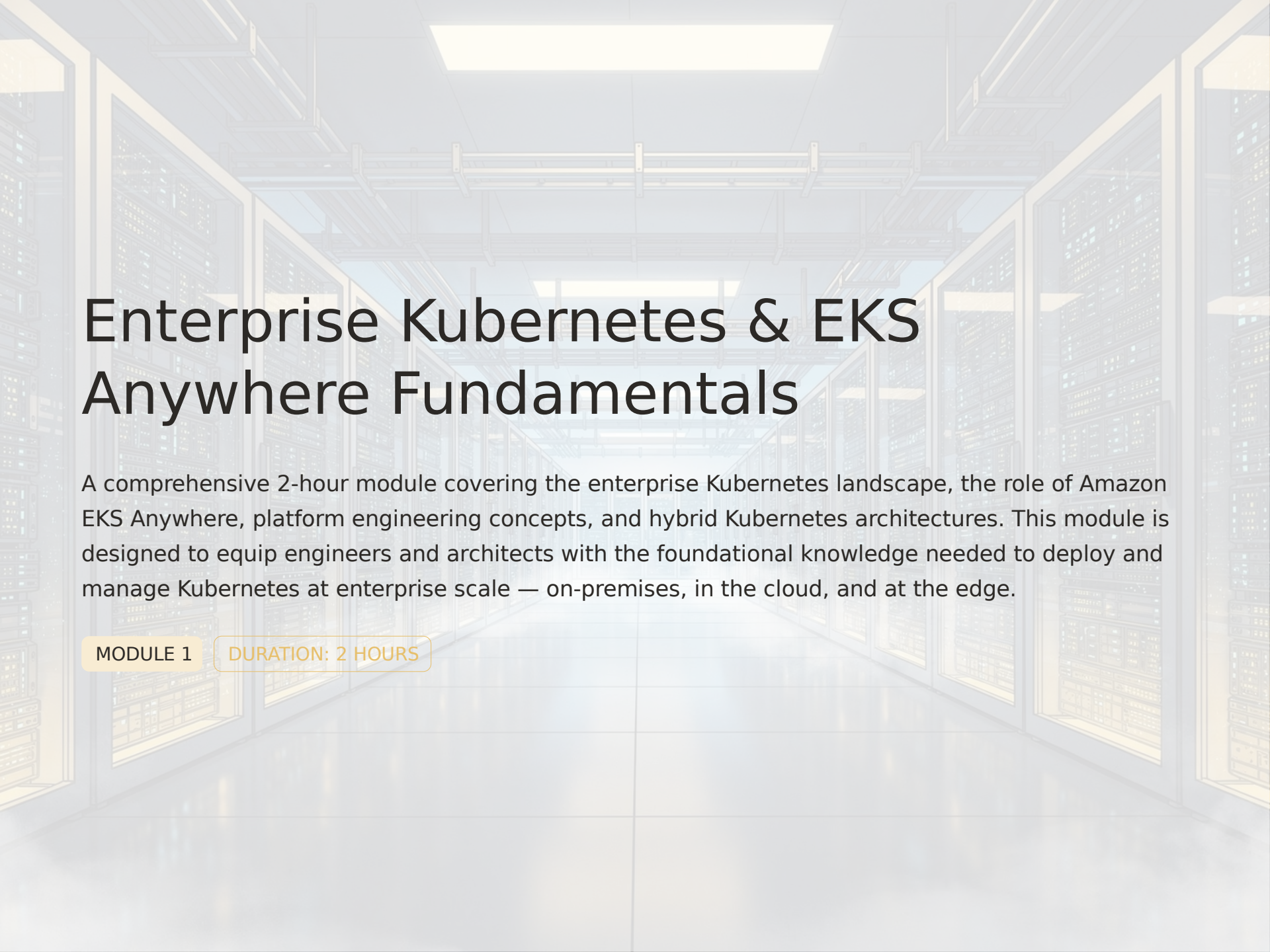




Enterprise Kubernetes & EKS Anywhere Fundamentals

A comprehensive 2-hour module covering the enterprise Kubernetes landscape, the role of Amazon EKS Anywhere, platform engineering concepts, and hybrid Kubernetes architectures. This module is designed to equip engineers and architects with the foundational knowledge needed to deploy and manage Kubernetes at enterprise scale — on-premises, in the cloud, and at the edge.

MODULE 1

DURATION: 2 HOURS

Enterprise Kubernetes Landscape

Modern enterprises have undergone a dramatic infrastructure transformation. Where organizations once relied on physical servers, virtual machines, manual deployments, and long release cycles, today's leading enterprises have embraced containers, Kubernetes, DevOps, GitOps, platform engineering, and infrastructure as code. This shift is not merely technological — it represents a fundamental change in how software is built, deployed, and operated at scale.

Why Enterprises Adopt Kubernetes

- Standard deployment platform across all environments
- Application portability between clouds and on-prem
- Self-healing applications with automated recovery
- Automated scaling based on demand
- Multi-cloud and vendor-neutral ecosystem support

Modern Enterprise Environments

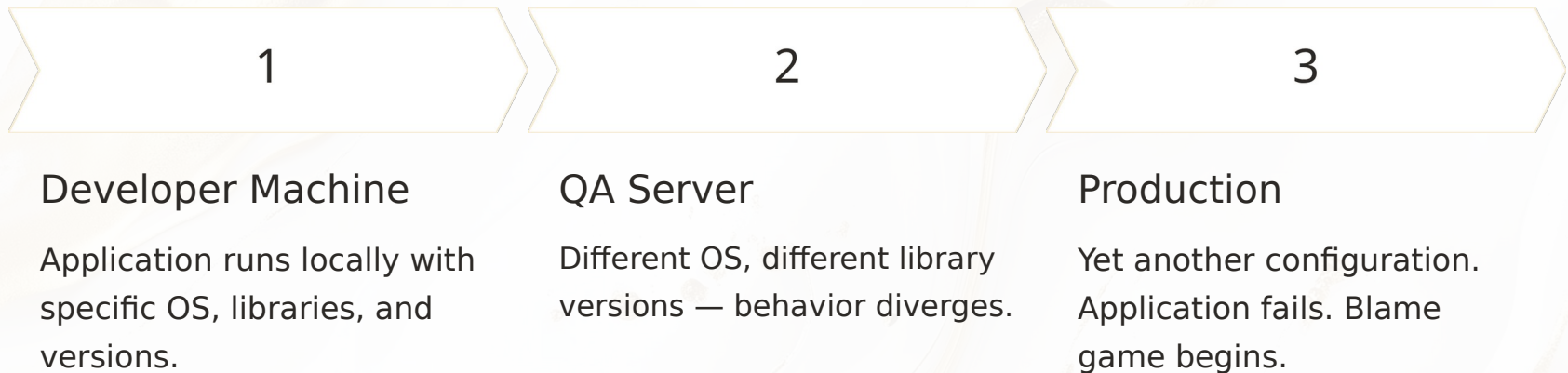
Ask yourself: how many organizations today run only one cloud? The answer is almost always none. Modern enterprises simultaneously operate across:

- AWS and Azure public clouds
- VMware vSphere on-premises
- Bare metal servers
- Edge devices and remote locations

Kubernetes provides a **common platform** across all of these environments, enabling consistent operations regardless of where workloads run.

Enterprise Challenges Before Kubernetes

Before Kubernetes, enterprises faced a persistent and costly set of operational problems rooted in inconsistency. Deployment processes differed across every environment, server configuration was largely manual, and dependency conflicts were a constant source of failure. The most damaging symptom of this era was environment mismatch — the dreaded scenario where an application worked perfectly on a developer's machine but failed in QA or production.



This environment mismatch problem was not an edge case — it was the norm. Teams spent enormous amounts of time debugging issues that were purely environmental, not logical. The lack of a consistent, portable runtime made reliable software delivery nearly impossible at scale.

How Kubernetes Solves the Problem

Kubernetes addresses the core challenges of pre-container infrastructure by introducing a consistent, portable, and automated runtime environment. By packaging applications and their dependencies into containers, Kubernetes guarantees that what runs in development is exactly what runs in production — eliminating environment mismatch at its root.



Same Container Everywhere

Identical runtime from dev to production — no more "works on my machine."



Predictable Deployment

Declarative configuration ensures consistent, repeatable deployments every time.



Automation

Scheduling, rollouts, and rollbacks are fully automated by the control plane.



Scalability

Horizontal pod autoscaling responds dynamically to real-time demand signals.

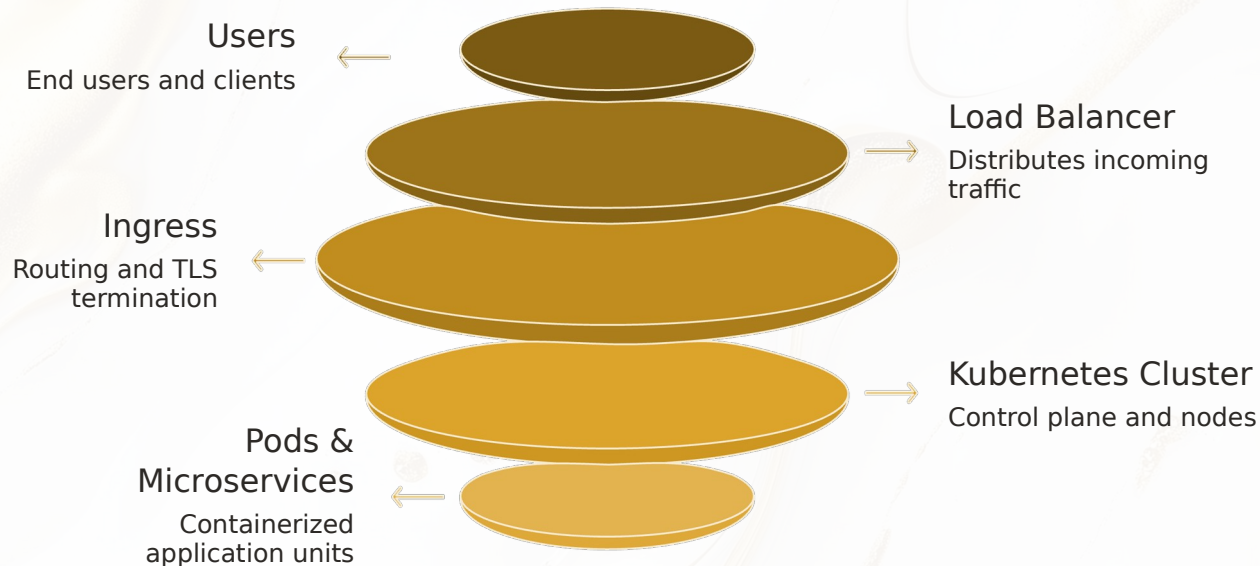


Self-Healing Recovery

Failed pods are automatically restarted, rescheduled, and replaced without manual intervention.

Kubernetes in Enterprise Architecture

In a production enterprise environment, Kubernetes sits at the heart of the application delivery stack. Traffic flows from end users through a load balancer, into an ingress controller, and then into the Kubernetes cluster where pods run the actual microservices. Those microservices connect to persistent databases and stateful backends. This layered architecture enables fine-grained traffic control, resilience, and observability at every tier.



Beyond the core request path, enterprise Kubernetes deployments are surrounded by a rich ecosystem of supporting services. Monitoring and logging provide full-stack observability. Security tooling enforces policy at the cluster and workload level. A service mesh manages inter-service communication with mutual TLS and traffic shaping. Backup and disaster recovery solutions protect stateful workloads and ensure business continuity in the event of cluster failure.

Why EKS Anywhere?

Amazon EKS delivers a fully managed Kubernetes experience in the AWS cloud — but not every workload can or should run in the public cloud. Regulated industries, sensitive environments, and latency-critical operations all have requirements that make cloud-only deployments impractical or impossible. Organizations in these sectors need the power and consistency of Kubernetes running on their own infrastructure.

Financial Services

Banks and financial institutions must keep sensitive data within their own controlled environments to meet regulatory requirements.

Healthcare

Hospitals handling patient data face strict compliance mandates that restrict data from leaving on-premises systems.

Manufacturing & Defense

Factory floors, military installations, and government facilities often operate in air-gapped or restricted-network environments.

This is precisely where **Amazon EKS Anywhere** enters the picture — bringing the EKS experience to on-premises and edge environments where the cloud simply cannot reach.

What is EKS Anywhere?

Amazon EKS Anywhere is an enterprise-grade Kubernetes distribution that allows organizations to run the full Amazon EKS experience entirely on their own infrastructure — without any dependency on the AWS cloud. It delivers the same Kubernetes APIs, the same CLI tooling, and the same operational lifecycle that teams already know from EKS, but deployed on infrastructure the organization owns and controls.



VMware vSphere

Deploy EKS clusters on existing VMware infrastructure, reusing current investments.



Bare Metal

Run directly on physical servers for maximum performance and control.



Nutanix

Leverage hyper-converged infrastructure with full EKS compatibility.



Snow & CloudStack

Extend Kubernetes to disconnected edge locations and private cloud platforms.



EKS Anywhere delivers the same Kubernetes APIs, same CLI, same lifecycle management, and enterprise support — on infrastructure you control.

Why Organizations Choose EKS Anywhere

EKS Anywhere addresses four distinct enterprise needs that public cloud alone cannot satisfy. Each use case represents a real-world constraint that drives organizations to deploy Kubernetes on-premises or at the edge, while still benefiting from the consistency and tooling of the EKS ecosystem.

Compliance & Data Sovereignty

Many organizations operate under regulations that prohibit sensitive data from leaving their own infrastructure. EKS Anywhere keeps data inside the organization's perimeter while still delivering modern Kubernetes capabilities.

Low Latency Requirements

Manufacturing lines, IoT sensor networks, and retail point-of-sale systems cannot tolerate the round-trip latency of a cloud API call. Running Kubernetes locally eliminates that bottleneck entirely.

Offline & Disconnected Environments

Ships at sea, defense installations, and oil rigs operate in environments with intermittent or no internet connectivity. EKS Anywhere runs fully air-gapped, ensuring operations continue regardless of network availability.

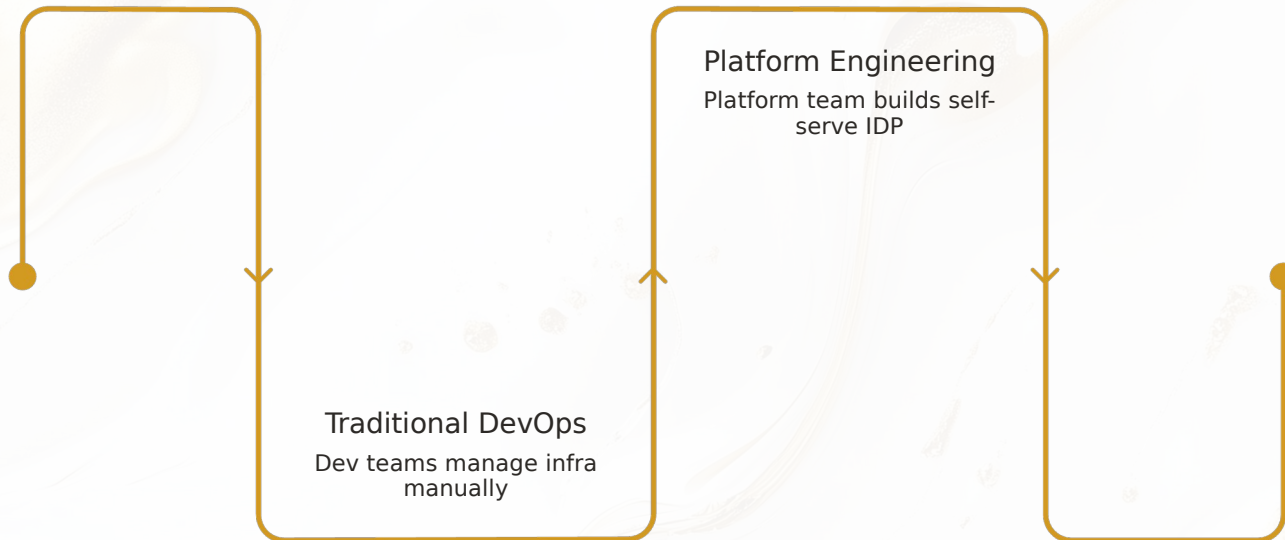
Existing VMware Investment

Organizations already running VMware vSphere can deploy EKS Anywhere directly on top of their existing infrastructure — no rip-and-replace required. This protects capital investment while modernizing the application platform.

Platform Engineering Concepts

Traditional DevOps placed the burden of infrastructure management squarely on individual development teams. Every team was responsible for provisioning, configuring, and maintaining their own pipelines, tools, and environments. While this model offered flexibility, it created enormous duplication of effort, inconsistent security postures, and a steep learning curve for developers whose primary focus should be writing application code.

Platform Engineering emerged as the answer to this scaling problem. A dedicated platform team takes ownership of the underlying infrastructure and tooling, building an **Internal Developer Platform (IDP)** that developers consume as a self-service resource. Rather than each team reinventing the wheel, the platform team creates golden templates, enforces security standards, and provides standardized pipelines that any team can use immediately.



The result is a dramatic improvement in developer experience. Engineers focus on building features rather than managing infrastructure. Security and compliance are baked into the platform by default. And the organization gains a consistent, auditable foundation for all software delivery.

DevOps vs. Platform Engineering

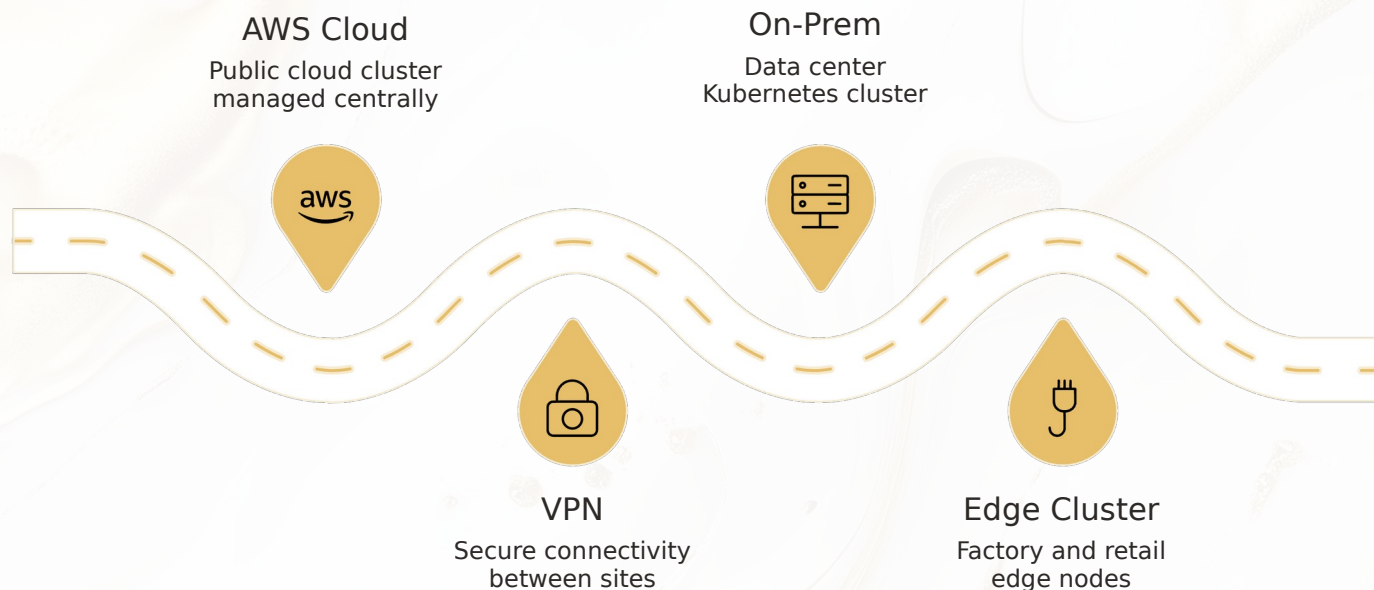
While DevOps and Platform Engineering share the goal of faster, more reliable software delivery, they differ fundamentally in how responsibility is structured, how infrastructure is managed, and what the developer experience looks like day-to-day. Understanding this distinction is critical for organizations designing their engineering operating model.

DevOps	Platform Engineering
Each team owns and manages its own infrastructure	Dedicated platform team owns shared infrastructure
Manual setup and configuration per team	Fully automated provisioning via self-service platform
Different pipelines across every team	Standardized pipelines shared organization-wide
Custom, fragmented tooling per project	Shared, curated toolset maintained centrally
High learning curve for new developers	Easy, consistent developer experience from day one

Platform Engineering does not replace DevOps culture — it operationalizes it. The collaboration, shared ownership, and continuous improvement mindset of DevOps remain essential. Platform Engineering simply adds structure and scale by centralizing the infrastructure expertise that was previously distributed and duplicated across every team.

What is Hybrid Kubernetes?

Hybrid Kubernetes refers to the practice of running Kubernetes clusters consistently across multiple environments simultaneously — public cloud, on-premises data centers, and edge locations — all managed through a unified control plane and operational model. Rather than treating each environment as a separate silo with its own tooling and processes, hybrid Kubernetes creates a single, coherent platform that spans the entire infrastructure estate.



The power of hybrid Kubernetes lies in consistency. The same deployment manifests, the same security policies, the same monitoring dashboards, and the same operational runbooks apply regardless of whether a workload is running in an AWS region, a corporate data center, or a cluster sitting on a factory floor. This dramatically reduces operational complexity and enables organizations to place workloads exactly where they need to be — based on latency, compliance, or connectivity requirements — without sacrificing manageability.

Hybrid Kubernetes Use Cases

Hybrid Kubernetes is not a theoretical architecture — it is actively deployed across a wide range of industries, each with distinct requirements that make a purely cloud-based approach insufficient. These real-world use cases illustrate why the ability to run Kubernetes consistently across cloud, on-premises, and edge environments is a strategic enterprise capability.



Retail

In-store Kubernetes clusters power point-of-sale systems, inventory management, and personalized customer experiences — even when internet connectivity is unreliable.



Manufacturing

Factory automation workloads require ultra-low latency and deterministic response times that only local edge clusters can provide.



Healthcare

Hospital applications handling patient data must remain on-premises to satisfy HIPAA and regional data sovereignty regulations.



Telecom & Finance

5G edge computing and regulatory financial workloads both demand infrastructure that combines cloud-scale tooling with on-premises data control — the defining promise of hybrid Kubernetes.

- 📌 Defense and air-gapped clusters represent the most demanding hybrid Kubernetes deployments — fully disconnected from public networks yet managed with the same EKS tooling.